From Steve to Spam — A Probabilistic Journey

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1. Introduction

Every day, our phone receives dozens of messages. Some are from friends, some are from your bank, and some are trying to sell you something you never asked for. Somewhere in our phone, a small piece of software reads each message and makes a quiet decision — spam or not spam.

That software is not following a list of rules. Nobody told it that messages containing the word "free" are suspicious. It figured that out on its own, using nothing but probability.

This report is about how that works. We will start with a simple puzzle about a man named Steve — a puzzle that most people get wrong — and use it to discover one of the most important ideas in probability. Then we will show how that exact same idea runs inside a real spam classifier, built from scratch, that correctly identifies 97.55% of messages it has never seen before.

2. A Puzzle About Steve

Read this description carefully:

> *"Steve is very shy and withdrawn, invariably helpful but with little interest in people or in the world of reality. A meek and tidy soul, he has a need for order and structure, and a passion for detail."*

Is Steve more likely to be a librarian or a farmer?

Take a moment. Most people say librarian immediately — shy, tidy, detail-oriented, loves order. It sounds exactly right.

But it is almost certainly wrong. And the reason why reveals something fundamental about how probability actually works.

2.1 The Question Nobody Asks

When most people read Steve's description, they ask: *does this match a librarian?* The answer is yes, strongly. So they conclude: librarian.

But there is a question they forget to ask first:

How many librarians and farmers actually exist?

In the United States, there are roughly 100,000 librarians and 2,000,000 farmers. That is 20 farmers for every 1 librarian. Before reading a single word about Steve, the base odds already say there is a 95% chance he is a farmer — just because farmers are so much more common.

Now bring in the description. Suppose 40% of librarians match Steve's personality. And suppose 10% of farmers do too — some farmers are

quiet and detail-oriented. So Steve's description does match librarians better. But watch what happens when we combine both pieces of information:

Librarian match: 1 × 0.40 = 0.40

Farmer match: 20 × 0.10 = 2.00

Out of a total of 2.40, the farmer's share is 2.00 — which is **83%**.

Steve is most likely a farmer. The prior — how common each group is — completely changed the answer.

2.2 What Just Happened

We just did something very specific. We started with a prior belief — farmers are 20 times more common. We observed evidence — Steve's description matches librarians better. We combined them — and got an updated belief that was different from both.This process of starting with what you know, observing evidence, and updating your belief has a name.

This is **Bayes' Theorem.**

$$P(\text{hypothesis} \mid \text{evidence}) = \frac{P(\text{evidence} \mid \text{hypothesis}) \times P(\text{hypothesis})}{P(\text{evidence})}$$

In plain words: your updated belief = how well the evidence fits × how likely the hypothesis was to begin with.

The insight that most people miss — and that Steve's example makes impossible to ignore — is that the prior is not optional. Forgetting it is not a minor error. It gives you the completely wrong answer.

3. The Spam Classifier — Same Idea, Bigger Scale

Now here is the interesting part. The question we just asked about Steve:

> *"Given this description, is he more likely to be a librarian or a farmer?"*

is exactly the question a spam classifier asks about every message:

> *"Given these words, is this message more likely to be spam or ham?"*

3.1 What the Classifier Does

(I have attached the complete code of this classifier model in a separate file, in the same repo, and here I have explained some important parts of the code.)

The classifier is trained on 5,000 labelled SMS messages from the UCI Spam Collection dataset. It reads them, counts everything, and builds a simple picture of the world:

```
"free" → appeared in 300 spam, 20 ham messages
"win" → appeared in 250 spam, 5 ham messages
"ok" → appeared in 8 spam, 1200 ham messages
"hello" → appeared in 10 spam, 800 ham messages
```

From this it computes two things. First, the prior — how common is spam overall?

P(spam) = 673 / 5000 = 0.1346

P(ham) = 4327 / 5000 = 0.8654

Only 13% of messages are spam. So before reading a single word, the classifier already leans toward ham — just like before reading Steve's description, we should already have leaned toward farmer.

Second, the likelihood — how characteristic is each word of each class?

We define functions to calculate the likelihood of a word appearing in a message given its class (spam or ham) using Laplace smoothing to handle words not present in the training set. We also define a function to calculate the likelihood of an entire document (message) given its class based on the presence or absence of words in the vocabulary.

```
alpha = 0.00001

def word_likelihood(term, category):
    if term in trainingSet:
        return (trainingSet[term][category] + alpha) / (count[category] + 2 * alpha)
    else:
        return alpha / (count[category] + 2 * alpha)

# Binary likelihood for a document
# doc_terms: list of words in the document

def doc_likelihood(doc_terms, category):
    def present(doc_terms, term):
        return 1 if term in doc_terms else 0
    likelihood_product = 1
    for vocab_word in vlist:
        prob = word_likelihood(vocab_word, category)
        likelihood_product *= (prob ** present(doc_terms, vocab_word)) * ((1 - prob) ** (1 - present(doc_terms, vocab_word)))
    return likelihood_product
```

The small value alpha prevents any word from getting a probability of exactly zero — because one unseen word would otherwise collapse the entire calculation.

3.2 Classifying a New Message:

When a new message arrives — say **"win free cash"** — the classifier computes a score for both spam and ham. It starts with the log of the prior and adds the log likelihood of each word:

We define functions to calculate the posterior score for a message belonging to each class using the calculated prior probabilities and word likelihoods. We also define a classification function that predicts the class of a message based on the posterior scores.

```
def posterior_score(doc_terms, category):
    score = math.log(class_prior(category))
    for term in doc_terms:
        score += math.log(word_likelihood(term, category))
    return score

def classify(words):
    if posterior_score(words, 'ham') > posterior_score(words, 'spam'):
        return "ham"
    else:
        return "spam"
```

We use logs to avoid numerical underflow — multiplying hundreds of tiny decimals causes the product to round to zero. Adding logs is mathematically equivalent and avoids this problem.

Watch how the scores evolve as we read the message word by word:

```
                    Spam Score    Ham Score    Leader
──────────────────────────────────────────────────────
Start (prior only):   -2.004       -0.144      Ham
After "win":          -2.994       -7.044      Spam
After "free":         -3.804      -12.344      Spam
After "cash":         -5.014      -18.564      Spam → FINAL ANSWER
```

The prior said ham. But the words were so strongly associated with spam that each one shifted the probability further and further — until ham was not even close. The classifier returns spam.

This is Bayes' Theorem being applied three times in a row — once per word. Each word is a new piece of evidence updating the belief.

4. Steve vs Spam — Two Sides of the Same Coin

At this point we might notice something. In Steve's example the prior won — even strong evidence couldn't overcome the fact that farmers are 20x more common. In the spam example the evidence won — even though 87% of messages are ham, three spammy words flipped the answer completely.

These are not two different theorems. They are the same theorem in two different situations.

Bayes' Theorem is a balance between prior and evidence. Sometimes the prior is heavy enough to dominate. Sometimes the evidence is strong

enough to override it. The theorem tells you exactly how much weight to give each — and the answer changes depending on the numbers.

Steve: Prior very strong (20x more farmers) → Prior wins → Intuition wrong

Spam: Evidence very strong (words 50x more common in spam) → Evidence wins → Intuition right

What they share is the same underlying question: *given everything I know, what is the most probable explanation?*

5. Results

Tested on 572 messages it had never seen, the classifier achieved:

Accuracy: 97.55% — 558 out of 572 correct.

Finally, we evaluate the performance of the classifier on the entire test set and display the accuracy and confusion matrix.

```
correct = 0
conf_matrix = {"ham-ham": 0, "ham-spam": 0, "spam-spam": 0, "spam-ham": 0}
for actual, words in testSet:
    guess = classify(words)
    correct += actual == guess
    conf_matrix[f"{actual}-{guess}"] += 1

print("Test accuracy:", correct / len(testSet))
print("Confusion matrix:", conf_matrix)
```

```
Test accuracy: 0.9755244755244755
Confusion matrix: {'ham-ham': 487, 'ham-spam': 11, 'spam-spam': 71, 'spam-ham': 3}
```

Only 14 messages were misclassified. The classifier caught 71 out of 74 spam messages — missing just 3. And it flagged only 11 legitimate messages as spam out of 498.

6. Conclusion

We started with a puzzle that most people get wrong. Steve sounds like a librarian. But once you ask how many librarians and farmers exist in the world, the answer flips completely. The prior matters — and Bayes' Theorem is the tool that makes sure you never forget it.

We then showed that the same reasoning drives a real spam classifier. The prior tells the model that most messages are legitimate. The likelihood of each word updates that belief. The posterior makes the final call.

Probability does not just describe uncertainty. Used correctly, it resolves it.